

**SYNTACTIC AND PROSODIC MARKING OF SUBJECT FOCUS IN AMERICAN
ENGLISH AND PENINSULAR SPANISH**

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Keywords: Information Structure, Syntax, Prosody, Pragmatics

Abstract

The aim of the present study is to provide an account of the different strategies, both syntactic and prosodic, employed by American English and Peninsular Spanish speakers in subject focus marking. Data obtained through parallel experimental designs revealed that prosodic marking of focus in-situ is possible in both languages both for informational and contrastive focus. Nonetheless, in the expression of contrastive focus Peninsular Spanish speakers increase the use of clefting while American English speakers exploit prosodic strategies like creaky voice. Differences in the pitch range implemented on focalized subjects were against the posed prediction. This study, nonetheless, contributes to the lacking cross-linguistic comparisons of these two languages and explores the interconnections between syntax and prosody.

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1. Introduction

1.1. Information structure and the different types of focus

The study of *information structure* is concerned with the mechanisms used across languages to introduce information in discourse. In this regard, information may be *given* when it is considered to be part of the common ground, that is, shared by both hearer and speaker. Information is *new* when it is not part of the common ground and the speaker introduces it in discourse in an attempt to fill an information gap (Haliday and Hassan, 1967; Prince, 1981; Krifka 2008). The linguistic device used to introduce new information in discourse is *focus*. Following the principles put forth by Rooth (1985) in *Alternative Semantics*, Krifka (2008) defined focus as the indicator of “the presence of alternatives that are relevant for the interpretation of linguistic expressions” (p. 247). In an utterance, we can then identify the linguistic constituents that are focalized (and therefore introduce new information) and differentiate them from those that present given information, which may be referred to as the *background*. One of the most popular approaches to the interpretation of focus is based on the idea of *question-answer congruence* (Roberts, 1996). According to this proposal, the focused constituent needs to highlight the same set of alternatives that was originally evoked by a specific question, which may be explicit or implicit.

Further distinctions can also be established considering the size of the focused constituent. In this regard, focus can be *narrow*, if it affects one constituent or even a smaller

unit, or *broad*, if it affects more than one constituent or even the entire utterance (Selkirk, 1984; Gussenhoven, 2008). Thus, an utterance with broad focus would answer the question ‘*What happened?*’, as shown in example (1), while narrow focus is elicited with wh-questions including as well background information, as in (2).

1) What happened?

_F[Julia spilled the lemonade]

2) Who spilled the lemonade?

_F[Julia] spilled the lemonade

Languages have different mechanisms to mark focus (Gussenhoven, 2008): morphological (e.g. Japanese, Bulgarian, or Irish), syntactic (e.g. Lekeitio Basque or Wolof), or phonological. Among the different phonological strategies, there are different possibilities: the presence of a pitch accent (e.g. in West Germanic languages, focused material receives a pitch accent while non-focused material is deaccented); the use of a specific pitch accent (e.g. in European Portuguese, H+L* is the neutral accent and H*+L is the focal accent); or the manipulation of prosodic phrasing (e.g. Chichewa). Across languages, however, focused constituents receive are prosodically prominent (Jackendoff, 1972; Selkirk, 1984; Büring, 2010), although the correlates of this prominence will be language-specific.

A further distinction is concerned with the strength of the focus being conveyed. In this sense, focus may be *informational* or *contrastive* (Kiss, 2008; Gussenhoven, 2008; Krifka, 2008; Zimmerman & Onea, 2011). In contexts of informational focus (also referred to as presentational), the speaker is solely providing information he believes to be missing in the common ground, as shown above in example (2). With contrastive focus, nonetheless, the speaker’s intention is to change a presupposition already held by the interlocutor, as represented

in example (3), and is elicited, in most cases, as a correction to a previous statement (Krifka, 2008). Conventionally, contrastive focus is denoted with the use of capital letters.

3) Jasmine spilled the lemonade

_F[JULIA] spilled the lemonade

1.2. The realization of focus in English and in Spanish

English and Spanish display some similarities regarding the assignment of nuclear stress (or main prosodic prominence). The unmarked word order in both languages is SVO and the default position for nuclear stress is the nucleus of the most embedded constituent in the sentence (Zubizarreta, 1998). Thus, main prosodic prominence (represented from now on in **bold**) is found in the same position in utterances with broad focus, as shown in (4-9).

4) _F[My cousin stole a **wallet**]

5) _F[Mi primo robó una **cartera**]

6) My cousin _F[stole a **wallet**]

7) My cousin stole _F[a **wallet**]

8) Mi primo _F[robó una **cartera**]

9) Mi primo robó _F[una **cartera**]

When considering those cases in which nuclear stress does not fall by default within the focused constituent, that is, for example, in utterances displaying subject focus, more differences between English and Spanish can be found. In this context, different strategies seem to become relevant. Three of these will be discussed in the present study: prosodic marking in-situ, clefting and p-movement.

Prosodic marking in-situ. This strategy results from the combination of two mechanisms: prominence shift and anaphoric deaccentuation. By means of prominence-shift, nuclear stress is

moved to the head of the focused constituent and, as a result of anaphoric deaccentuation, the background information is produced without any prosodic prominence (Zubizarreta, 1998; Gussenhoven, 2004). Thus, in an utterance with subject focus such as the one presented above in (2) and repeated below in (10), main prominence falls on the subject. Its use is accepted for English both in the realization of informational and contrastive focus (Zubizarreta, 1998; Gussenhoven, 2004). For Spanish, however, the availability of this strategy is more controversial. Zubizarreta (1998) argues that an utterance with prosodic marking in-situ such as the one presented in (11) could only be uttered when conveying contrastive focus. Nonetheless, experimental evidence has shown that speakers of different varieties of Spanish, including Peninsular Spanish, can in fact mark informational focus in-situ by means of prosody (Gabriel 2010; Vanrell & Fernández-Soriano, 2013; Feldhausen & Vanrell, 2014). What still remains unclear, however, is whether post-focal material (that is, background information) is completely unaccented as a result of *post-focal deaccentuation* or whether a different mechanism, referred to as *post-focal compression*, applies (Nadeu & Vanrell, 2015; Vanrell & Fernández-Soriano, in press). Post-focal compression is characterized by the use of pitch accents with lower mean and maximum F0, and a more compressed pitch range.

10) Who spilled the lemonade?

F[**Julia**] spilled the lemonade

11) F[**Julia**] derramó la limonada

Clefting. This strategy involves the realization of focalized material in a cleft introduced by a conjugated form of the verb *to be* (*ser* in Spanish) while background information is expressed in a relative clause (Moreno Cabrera, 1999; Lambretch, 2001), as shown in (12) and (13). This strategy is available in both languages but favored in contexts of contrastive focus, as

predicted by Féry (2013) in her *Focus as Alignment* model. In summary, she proposes that as a result of being a “stronger” type of focus than informational focus, contrastive focus is more likely to display additional prosodic cues. By means of clefting, the focused constituent is not only the most prominence constituent within the utterance but is also aligned with the right edge of an intonational phrase, which makes it more prominent.

12) It was _F[**Julia**] who/that spilled the lemonade

13) Fue _F[**Julia**] quien derramó la limonada

P-movement. Zubizarreta (1998) claims that, instead of prosodic marking in-situ, *prosodically motivated movement* (p-movement) is employed in Spanish to convey informational focus in a position that does not receive nuclear stress by default. In short, this strategy consists of the scrambling of the constituent(s) introducing given information, which then results in the placement of the focused constituent in final position, where prosodic prominence is automatically assigned as shown in (14). Experimental evidence for the use of this strategy has been found mostly for Peninsular Spanish (Feldhausen & Vanrell, 2014).

14) Derramó la limonada _F[**Julia**]

Focus can be prosodically marked in-situ both in English and in Spanish, although the correlates of focal prominence need not be the same. To the best of our knowledge, there are no empirical studies comparing the realization of focus in American English and Peninsular Spanish. Nonetheless, a preliminary description of how these two languages may differ can be proposed on the basis of the descriptions of their intonational grammars, framed within the Autosegmental-Metrical framework, and the findings from several studies that have compared British English and Peninsular Spanish.

The Autosegmental-Metrical (AM) framework (Pierrehumbert, 1980; Beckman & Pierrehumbert, 1988; Ladd, 1996) and the Tones and Break Indices (ToBI) systems developed afterwards for the annotation of intonation have facilitated the description of intonation across languages. The ToBI system for Mainstream American English (MAE_ToBI) was developed by Beckman and Hirschberg (1994) and Beckman, Hirschberg, and Shattuck-Hufnagel (2005), among others. The ToBI system for Spanish (Sp_ToBI) was proposed by Beckman, Díaz-Campos, McGory, and Morgan (2002) and further developed by Estebas-Vilaplana and Prieto (2009), Aguilar, de la Mota, and Prieto (2009), Prieto and Roseano (2010), and Hualde and Prieto (2015).

Even though ToBI systems are language-specific, which implies that the labels employed do not necessarily describe exactly the same tonal movements, it is still possible to draw some assumptions about how American English and Peninsular Spanish differ in terms of the intonational implementation of focus. The label proposed for American English in contexts of broad focus and informational focus is H*, which would contrast with L+H*, found in contexts of contrastive focus (Pierrehumbert & Hirschberg, 1990). In Spanish, the pitch accent that is generally found in pre-nuclear position in unmarked contexts (broad focus) is a rising pitch accent with a late peak, represented with the label L+<H* (Hualde & Prieto, 2015). In contexts of contrastive focus, the use of L+H* is generally accepted for most Spanish varieties (Gabriel, 2010; Estebas-Vilaplana & Prieto, 2009; Vanrell, Stella, Fivela, & Prieto, 2013; Hualde & Prieto, 2015) and is usually followed by a L- boundary tone when produced in pre-nuclear position (Vanrell & Fernández-Soriano, in press), although some variation was found by Face (2001). In contexts of informational focus, the use of the configuration L+H* L- has been reported (Vanrell & Fernández-Soriano, in press), although studies with a bigger pool of

participants have found more variation in this regard, suggesting that the differences between broad and informational focus cannot be described through pitch categories (Face, 2002); instead, more attention needs to be paid to other prosodic parameters (pitch range, alignment and duration). In conclusion, what the labels proposed for these two languages suggest is that while English speakers exploit differences in pitch range (H^* vs. $L+H^*$), Spanish speakers tend to manipulate alignment ($L+<H^*$ vs. $L+H^*$). In fact, cross-linguistic comparisons between British English and Peninsular Spanish (García-Lecumberri, 1995), as well as studies on the L2 acquisition of English focus marking by Spanish native speakers (Ramírez Verdugo, 2006; Ortega-Llebaria & Colantoni, 2014) suggest that English speakers employ a wider pitch range. The analysis carried out in this study aims to determine whether differences in pitch range are relevant when considering both contrastive and informational focus in the comparison of American English and Peninsular Spanish.

A further strategy that needs to be taken into consideration is the realization of post-focal material. As described above, the use of deaccentuation is generally accepted for West Germanic languages like English. Some of the correlates of deaccentuation are lower mean F_0 , narrower pitch range, faster speech rate and phonetic hypoarticulation (Johnson, 2003; Ortega-Llebaria & Prieto, 2011). For Spanish, nonetheless, Nadeu and Vanrell (2015) suggest that, due to the lack of complete deaccentuation, the most appropriate way to describe the realization of post-focal material is through the term “post-focal compression”. Further insight will be provided in the present analysis with regards to this issue. In particular, the role of creaky voice will be explored as a possible additional cue to convey background information. Non-constricted creaky voice is associated with low and irregular F_0 (Lindblom, 2009) and high H_1 - H_2 values, that is, increased difference in the amplitude found in the first and second harmonics (Keating, Garellek, &

Kreiman, 2015). To the best of our knowledge, this is the first study to perform an analysis of creakiness in a comparison of the realization of focus in English and Spanish.

2. Present study

The goal of this study is to provide an account of the different strategies, both syntactic and prosodic, employed by speakers of Mainstream American English (MAE) and Peninsular Spanish (PS) to mark subject focus. Regarding the prosodic realization of focus, special attention will be paid to the role of F0 and pitch range. To achieve this goal, two research questions will guide the design of the study and the analysis of the data obtained:

1. Do MAE and PS speakers use prosodic marking in-situ to the same extent?
2. When they do use this strategy, how do these two languages differ in terms of the prosodic realization of focus?

The hypothesis for the first question is that even though both groups will use prosodic marking of focus in-situ, MAE speakers will use it more often than PS speakers, who will employ a variety of strategies, including clefting and p-movement. Furthermore, following Féry's (2013) model, it is expected that the use of clefting will increase in both languages in contexts of contrastive focus.

Regarding the second research question, the hypothesis is that MAE speakers will use a wider pitch range on contrastively focused subjects than in any other context. This focal accent will also display a wider pitch range in MAE than in PS. Pitch range, however, will not be relevant for PS speakers. Furthermore, the realization of post-focal material will be analyzed as well in order to provide a more detailed answer to the second research question. In this regard, it is predicted that MAE speakers will deaccent linguistic material conveying background information, which will result in narrower pitch range, low F0 values and creaky voice.

Utterances produced with subject focus in PS, on the other hand, will display post-focal compression, that is, pitch accents in the constituent conveying background information (the verbal phrase or VP) will be produced with a narrower pitch range throughout the VP when the focus is on the subject.

3. Methodology

3.1. Experimental materials

A contextualized *Questions & Answers* task eliciting semi-spontaneous speech was designed. Participants were asked to pretend they were having a video conversation with an old friend from high school who would ask them questions about their classmates' whereabouts in the present. The experiment was presented on a PowerPoint presentation and the information to answer each question was presented in the next slide in the form of posts from the relevant person's social media. An example of the different slides designed for one specific item is presented in *figure 1*.



Figure 1. The picture on the left displays a screenshot of the slide where the question was asked (in this case “I also heard that somebody lost their wallet with a lot of money in it. Do you know who?”). The picture on the right displays the slide that would immediately follow the video and present the information required to answer the question (the name of the person and the event).

All the target utterances featured transitive verbs and were elicited first with informational focus (IF) and then with contrastive focus (CF). Three different focus structures were included in the design: subject focus (SF), VP focus (VpF), and object focus (OF). A total of 8 utterances were elicited per condition, resulting then in a total of 48 target utterances per participant. Two versions of the experiment were created, one in English and another one in Spanish.

3.2. Procedures

Participants first completed a linguistic background questionnaire to provide information about their age, native language, linguistic knowledge, and linguistic practices (e.g. use of Spanish outside of the classroom). Then, after a warm-up activity, they read the instructions for the experimental task, completed a practice item and proceeded with the experimental task at their own pace. Throughout the session, participants were recorded using a *USB Logitech* microphone connected to a desktop or a laptop computer. Recording sessions were carried out in a lab environment and digitized at a 44,100 Hz sample rate and a 16 bit amplitude resolution.

3.3. Participants

The experiment was presented to five speakers of MAE and five speakers of PS. They were all native speakers of their respective language and only beginner or intermediate learners of a second language, as reported in the linguistic background questionnaire. MAE speakers were recruited from the New England area, and they were 22.2 years old on average (ranging from 20 to 25). PS speakers were all from the northern region of Asturias and they were 24.8 years old on average (ranging from 22 to 28).

3.4. Coding and analysis

A total of 480 utterances were obtained. These were first coded for the specific syntactic strategy used. For this purpose, three main categories were identified: *In-situ*, *clefting* and *other*. The category *other* includes p-movement, dislocations or responses that were not produced in a complete sentence.

Then, all the utterances produced with prosodic marking of focus in-situ were subjected to further phonetic analysis using the software *Praat* (Boersma & Weenink, 2015). *Table 1* provides information about the exact number of utterances in each condition. Using *Praat*, the following values were extracted from the NP functioning as the subject and from the VP: a) pitch range, calculated as the difference between maximum and minimum F0; b) the difference between the first and the second harmonics (H1-H2), which would then allow for the measurement of creakiness. F0 values were measured in semitones, as this scale (a logarithmic transformation of the Hertz scale) allows for the normalization of data (Nolan, 2003) across participants.

		English	Spanish
Informational Focus	Subject focus	66	50
	VP focus	70	74
	Object focus	72	74
Contrastive focus	Subject focus	74	34
	VP focus	74	78
	Object focus	80	72
Total		436	382

Table 1. *Distribution of items displaying prosodic marking in-situ*

The statistical analysis was performed using *RStudio* (R Core Team, 2014) and the package *lmerTest* (Kuznetsova, Brockhoff, & Christensen, 2016). To answer the first research question, concerned with the frequency of use of prosodic marking in-situ, the three syntactic categories that were originally identified in the data (prosodic marking in-situ, clefting, and other) were reduced to *prosodic marking in-situ* vs. *other*. Then, a logistic regression model was performed, taking this binary version of the factor *strategy* as the dependent variable; *language* (English vs. Spanish) and *focus type* (Informational vs. Contrastive) were taken as the independent variables. This model explored both the main effect of these two variables and the interaction between them. To answer the second research question, three linear regression models were performed. The dependent variables were: *pitch range values within the subject*, *pitch range values within the VP*, and *H1-H2 values within the VP*. The independent variables were: *language* (English vs. Spanish), *focus type* (Informational vs. Contrastive), and *focus structure* (subject focus, VP focus, and object focus). All these models explored the main effect of each of these variables as well as their interactions.

4. Results

4.1. Syntactic strategies

The three main categories for the syntactic realization of focus were prosodic marking in-situ, clefting, and other. *Figure 2* below presents the proportion of use of each of these strategies in the realization of subject focus in contexts of informational and contrastive focus.

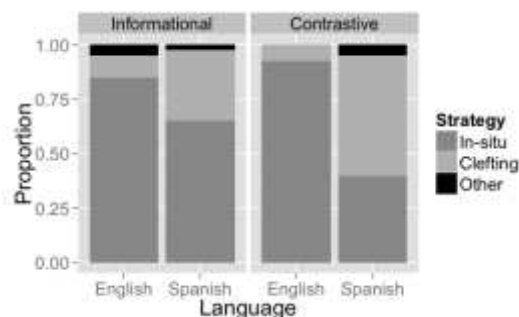


Figure 2. *Strategies used in the realization of subject focus*

For informational subject focus, prosodic marking in-situ was the preferred strategy, both in English and in Spanish. Nonetheless, its use was significantly more frequent in English than in Spanish ($\beta = 1.11$, $SE = .55$, $t = 2.01$, $p < 0.05$). In both languages, the second most common strategy was clefting, although this varied across participants.

In contexts of contrastive focus, English speakers used mostly prosodic marking in-situ. This strategy was consistently employed by four out of the five participants. When comparing both languages, Spanish speakers employed prosodic marking in-situ significantly less often than English speakers ($\beta = -2.91$, $SE = .68$, $t = -4.28$, $p < 0.001$), as it was the case with informational focus. Furthermore, the proportion of use of this strategy in contexts of contrastive focus by Spanish speakers decreased significantly in comparison with contexts of informational focus ($\beta = -1.02$, $SE = .46$, $t = -2.21$, $p < 0.05$); in other words, the use of clefting increases in contexts of contrastive focus.

4.2. Prosodic marking in-situ

In order to explore the prosodic implementation of subject focus, the pitch range manifested in the pitch accent realized within the subject was considered. *Figure 3* displays the pitch range values found in each condition for each language.

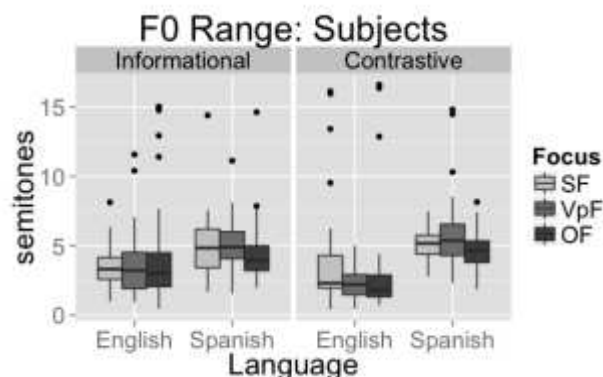


Figure 3. *Pitch range values within the subject.*

The results from the statistical analysis revealed that, for English, there are no differences in contexts of informational focus with regards to the pitch range produced on the subject; that is, the same pitch range is produced regardless of whether the subject is focused or not. Similar results were found for Spanish. Across conditions, the pitch range values reported for Spanish are higher than those reported for English. When considering specifically the realization of focused subjects, the results from the statistical analysis indicate that Spanish speakers actually produce a significantly wider pitch range than English speakers ($\beta = 1.49$, $SE = .7$, $t = 2.1$, $p < 0.05$). An example of how the intonational contours differ is shown in *figure 4*.

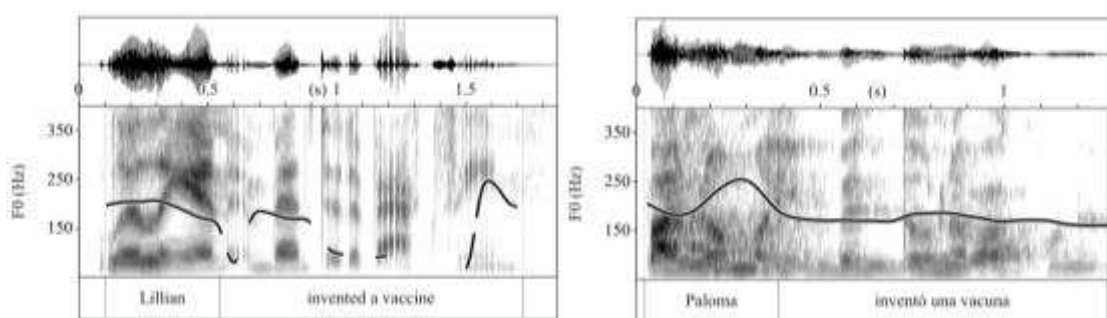


Figure 4. *Intonational contours found in utterances expressing informational subject focus in English (on the left) and Spanish (on the right)*

In contexts of contrastive focus, significant differences in the realization of focused and non-focused subjects were only found when looking at utterances with focus on the VP as produced by English speakers ($\beta = -1.61$, $SE = .62$, $t = -2.6$, $p < 0.01$), since these subjects tended to display a significantly narrower pitch range. Nonetheless, the pitch range values manifested in contrastively focused subjects were not significantly different from those found in contexts of informational focus for either language. Finally, the comparison between languages did not reveal, in this case, any significant differences, although the trend was similar to the one described for informational focus: Spanish speakers seem to be producing a wider pitch range, as shown in *figure 5*.

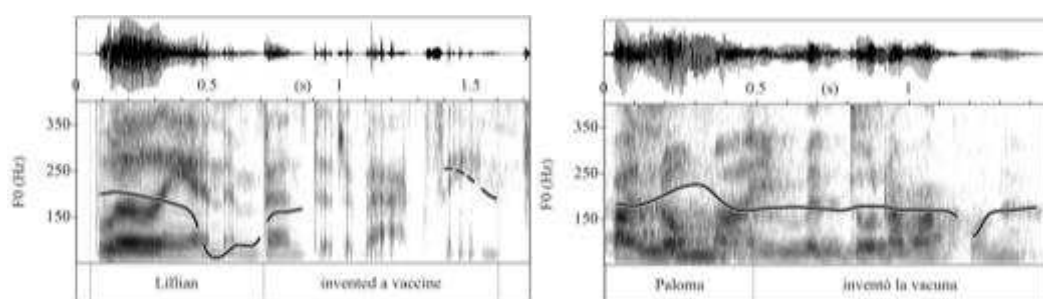


Figure 5. *Intonational contours found in utterances expressing contrastive subject focus in English (on the left) and Spanish (on the right)*

The realization of post-focal material was then further analyzed. *Figure 6* displays the pitch range values found within the VP in each focus condition.

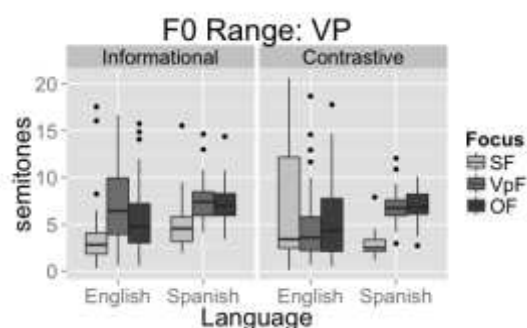


Figure 6. *Pitch range values (in semitones) within the VP.*

The statistical analysis revealed that, in both languages, the VP tends to be realized within a narrower pitch range when the subject is informationally focused. The difference was significant when comparing utterances with informational subject focus with those produced with VP focus in English ($\beta = -3.81$, $SE = .89$, $t = -4.28$, $p < 0.001$) and in Spanish ($\beta = -2.59$, $SE = .92$, $t = -2.81$, $p < 0.01$), as well as with utterances produced with object focus in Spanish ($\beta = -2.18$, $SE = .92$, $t = -3.27$, $p < 0.05$). No significant differences were found between English and Spanish when considering the pitch range displayed in the VP in contexts of informational subject focus. Furthermore, no significant differences were found when comparing informational and contrastive focus in Spanish, which indicates that the pitch range produced in utterances with contrastively focused subjects is as narrow as the one produced in utterances with informational focus. Nonetheless, when examining the same comparison for English, the results indicate that a wider pitch range was produced in contexts of contrastive subject focus ($\beta = 2.17$, $SE = .91$, $t = 2.38$, $p < 0.05$). A quick observation of the values displayed in *figure 6* reveals that a large amount of variation is found in this focus condition. Due to this variation, no differences were found across focus structures in the English data. For Spanish, on the other hand, the trends are similar to those reported for informational focus: utterances with focused elements within the VP display significantly higher pitch range values, both in contexts of VP focus ($\beta = 3.9$, $SE = 1.03$, $t = 3.77$, $p < 0.001$) and object focus ($\beta = 4.04$, $SE = 1.04$, $t = 3.86$, $p < 0.001$) than utterances with focalized subjects.

Further examination of the pitch range values found in the speech of English speakers when conveying contrastive focus revealed that the variation mainly resided in the minimum F0 values. Thus, English speakers were using extremely low pitch throughout the VP when this syntactic constituent did not include any focused word. This trend motivated the examination of

the amount of creak. *Figure 7* displays the H1-H2 values extracted from the VP for each focus condition in both languages.

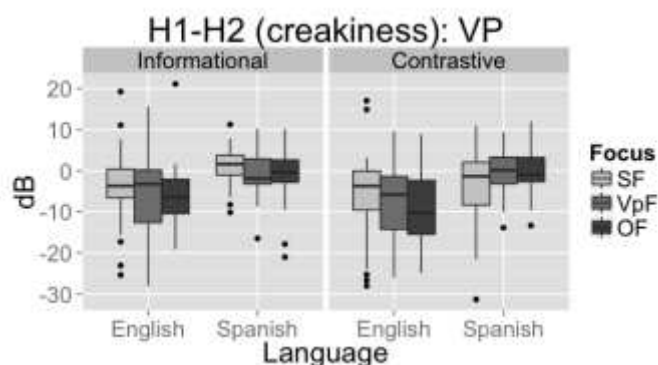


Figure 7. *H1-H2 values (measured in decibels) in the VP*

The results from the statistical analysis revealed that in utterances with contrastively focused subjects, H1-H2 values were significantly higher than in utterances where the object was contrastively focused ($\beta = 3.55$, $SE = 1.76$, $t = 2.01$, $p < 0.05$). Since non-constricted creaky voice is characterized by higher H1-H2 (Keating et al., 2015), the patterns found in this study do suggest that the use of creaky voice increases in the realization of background material. It is interesting to note that no differences were found between contrastive and informational focus, or when comparing solely utterances with informational focus, although the trend seems to be parallel, as shown in *figure 7*. No significant differences were found when examining the Spanish data.

The examples presented in *figure 8* show how the English and the Spanish intonational patterns differ in utterances expressing informational subject focus. In English, the stressed syllable in the subject is realized with a high tone; immediately after, the F0 falls and becomes more irregular, indicating as well the presence of creak, which can be identified as well in the spectrogram by examining the glottal pulses throughout the word *ballads* (circled). In Spanish, on the other hand, a rising tonal movement is realized within the stressed syllable and, even

though some tonal movement is still occurring within the VP, its pitch range is much narrower than what we could expect in a neutral declarative.

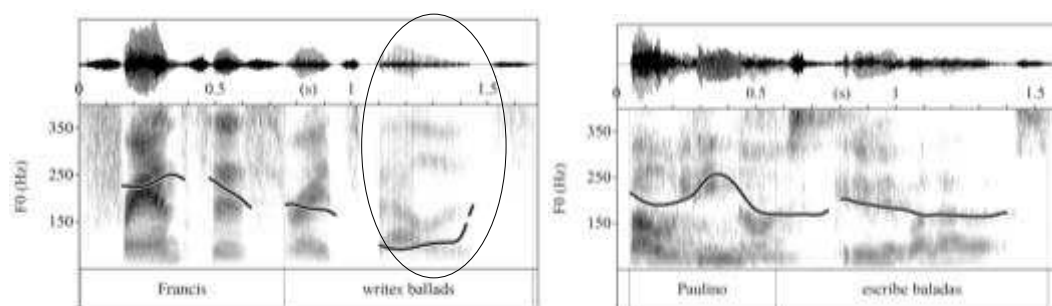


Figure 8. *Intonational contours found in utterances expressing informational subject focus in English (on the left) and Spanish (on the right)*

5. Discussion

In light of the results presented, the research questions and their respective hypotheses will be discussed. In particular, the first research question was concerned with the frequency of use of prosodic marking in-situ in MAE and PS. The hypothesis was that prosodic marking in-situ would be the preferred strategy for MAE speakers while PS speakers would use a variety of strategies. The results indicated that the use of prosodic marking of focus in-situ is in fact significantly more frequent in MAE than in PS, both in contexts of informational and contrastive focus. No significant differences between contexts were found in MAE, that is, MAE speakers mark focus through prosody almost consistently regardless of the strength of the focus being conveyed. PS speakers, on the other hand, increase the use of clefting in contexts of contrastive focus. These results provide further evidence in favor of the availability of prosodic marking in-situ in PS, not only in the expression of contrastive focus (Zubizarreta, 1998) but also in contexts of informational focus (Vanrell & Fernández-Soriano 2013, in press). Thus, different strategies are at the speakers' disposal, and there is no one-to-one mapping between a specific strategy and

the type of focus being expressed; as a result, we might expect optionality not only across dialects, but also within the same speakers, as argued also by Dufter and Gabriel (2016).

The second research question was concerned with the prosodic realization of focus when marked prosodically in-situ. Specifically, the role of pitch range was explored, as this prosodic feature had already been identified as a divergent cue when comparing (British) English and PS. The hypothesis was that MAE speakers would make use of a wider pitch range than PS speakers to mark contrastive subject focus. Nonetheless, this prediction was not confirmed. The comparison of the pitch range values found in the implementation of focal pitch accents in subject position revealed, instead, an opposite trend: PS speakers realized subjects within a wider pitch range than MAE speakers. Furthermore, there were no significant differences in pitch range between informational and contrastive focus, at least in terms of the pitch range implemented on the subject. Pitch range was not expected to play a particular role in PS, as alignment has been suggested to be the main cue. These results, however, are surprising for MAE, considering that a phonological contrast between H* and L+H* had been proposed (Pierrehumbert & Hirschberg, 1990). One possible explanation is that these findings provide further support for the argument that the phonological contrast between H* and L+H* is not real (Watson, Tanenhaus, & Gunlogson, 2008; Breen, Fedorenko, Wagner, & Gibson, 2010; Katz & Selkirk, 2011). Instead, the relative difference between focal and post-focal material would be the most relevant cue. This will be further discussed below. An alternative reason for the unexpected results is that they may have been affected by the accentual patterns and the phonetic structure of the stimuli employed. In the Spanish data, the stressed syllable of the subject was always in medial position. In English, however, stress always fell in initial position. Thus, it may have been the case that MAE speakers did not have enough phonetic material to actually implement a rising pitch

accent, which could then explain why lower pitch range values were reported. Future studies should then control for this variable in order to more strongly contribute to the debate on the phonological contrast between H* and L+H*.

The second research question was also concerned with the realization of post-focal material and, in particular, with the use of deaccentuation. The prediction in this regard was that MAE speakers would use more deaccentuation than PS speakers. The data obtained revealed that, in both languages, post-focal material in utterances with informational subject focus is produced within a narrower pitch range than in any other contexts. This was the case as well in utterances with contrastive focus as produced by PS speakers. In the case of MAE, nonetheless, the wide amount of variation found resulted in the lack of significant differences between contexts. This variation was due to the production of low and irregular F0. Since these features have been associated with the use of creaky voice, further analyses were performed on the data exploring, in particular, the differences in amplitude between the first and the second harmonics (H1-H2). In this regard, higher H1-H2 indicated that, in fact, MAE speakers were using creaky voice significantly more in utterances with contrastive subject focus than in any other context. Taken together, all these findings provide support for the hypothesis proposed. PS speakers use at least post-focal compression, if not deaccentuation, both in contexts of informational and contrastive focus. Nonetheless, considering in particular the realization of contrastive focus, it is also evident that MAE speakers are using additional strategies that do not seem to be available in PS, that is, an increased creak throughout the constituent conveying background information. This finding can in fact be related to the distribution of syntactic strategies reported above: Spanish speakers are not exploiting pitch range differently to mark contrastive focus but, instead, they display greater preference towards the use of syntactic strategies such as clefting, which will

make the focused constituent even more prominent. Thus, as predicted by Féry (2013), the strength of the focus being expressed will have an effect not only on the prosodic but also on the syntactic realization of focus. These preferences, nonetheless, are language-specific and seem to be closely related with the availability of specific prosodic mechanisms involved in the assignment of prominence.

There are other prosodic features that should be further explored, as they may be exploited differently across languages. In this regard, the role of peak alignment in Spanish has already been mentioned. Other prosodic parameters that have been reported to contribute to the realization of focus are duration and intensity. Furthermore, the approach of the analysis may differ as well. In the present study, we decided to examine the difference in pitch range values within each syntactic constituent, that is, the analysis of subjects and verbal phrases was independent but the relative difference in peak height between constituents could also be analyzed. Thus, future studies should explore the interplay between different prosodic parameters, controlling as much as possible for the phonetic structure of the target words (which may be particularly challenging in cross-linguistic comparisons), and considering the most appropriate approach.

6. Conclusions

This study constitutes, to the best of our knowledge, the first approach to an empirical comparison of the syntactic and prosodic strategies employed in American English and Peninsular Spanish to mark subject focus. Using a parallel experimental design, we have shown that, contrary to what has been previously proposed, prosodic marking in-situ is possible not only in English, but also in Spanish. Nonetheless, as it would be expected, the intonational and prosodic parameters employed to mark focus need not be the same.

This study, in particular, has provided more insight into the role of pitch range, a prosodic parameter that had already been identified as relevant for British English speakers compared to Peninsular Spanish speakers. However, no differences were found in terms of pitch range when comparing informational and contrastive focus for either language. Actually, it was found that Peninsular Spanish speakers produced informationally focused subjects within a wider pitch range than American English speakers, contrary to what had been predicted. While this may have resulted from the different accentual patterns and phonetic material of the subjects used in the elicited utterances, future studies should further explore this finding, as it could also constitute evidence that the difference in pitch range within the subject may not be as relevant in English as previously suggested. The realization of post-focal material, on the other hand, did reveal significant differences between the two languages under study: while both languages display compressed pitch range throughout the constituent conveying background, American English speakers use as well creaky voice in utterances with contrastive focus. This strategy seems to work as an additional cue to highlight the informational status of the subject, making it even more prominent than in utterances with informational focus. Peninsular Spanish speakers, nonetheless, did not seem to be using any of the prosodic parameters under consideration to distinguish informational from contrastive focus. Instead, the findings in this study suggest that in order to make the subject even more prominent, Spanish speakers seem to be more inclined to using syntactic strategies like clefting, which could be interpreted as a way to compensate for the lack of additional prosodic mechanisms to assign prominence.

Further studies are needed in order to better understand how the intonational grammars of these two languages differ, since an increased understanding of the strategies available in each

one of them will facilitate the study of how they interact with other components of the grammar and how these may affect language change phenomena as well as language learning.

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